

Ultrasonication outperforms electroporation for extracellular vesicle cargo depletion

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Abstract

Extracellular vesicles (EVs), submicron-sized membranous structures released by cells, serve as vehicles of tissue-specific proteins and nucleic acids, facilitating intercellular communication and playing roles in pathophysiological processes. Leveraging their unique characteristics, EVs have emerged as promising drug delivery nanocarriers. Electroporation (EP) and ultrasonication (US) are among the prevalent techniques used for loading exogenous drugs into EVs owing to their simplicity and efficiency. However, the effectiveness of the two methods in depleting initial EV cargo has been overlooked. But this information is indispensable, as the bioactive residuals of EVs, notably derived from tumor or stem donor cells, may impact downstream therapeutic effects. Bridging this knowledge gap, therefore, can guide the selection of optimal drugs and loading methods tailored to therapeutic objectives. Here, we used high-throughput sequencing to investigate the protein and small RNA cargo of EVs treated with EP and US, respectively. We found that US exhibits higher efficacy in depleting EV cargo compared to EP, while US may also deplete essential endogenous molecules for combination therapy. Neither method demonstrated significant selectivity in cargo depletion, but they might preferentially retain few specific molecules. Additionally, membrane proteins are more prone to loss during US and EP treatments than cytoplasmic proteins.